

CampusFind Pro

Complete Technical & Implementation Documentation

Created automatically via Antigravity Coding Assistant

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1. Project README Overview

CampusFind Pro ■

CampusFind Pro is a premium, modern, and secure lost-and-found portal designed specifically for university campuses. Built with PHP 8+, MongoDB Atlas, Google OAuth 2.0, Bootstrap 5, and integration for transactional email delivery via Brevo and Resend APIs.

■ Key Features

- Google OAuth Login: Quick, secure authentication mapped automatically to campus student profiles.
 - Lost & Found Reporting: Log detailed item reports with photos, categories, dates, and reward estimates.
 - Interactive Dashboard: View real-time personal statistics, pending/approved claims, notifications, and recent activity logs.
 - Administrative Control Panel: Manage users, edit/delete logs, approve claims, and customize system settings.
 - Email Notifications: Automatic dispatch of password reset links and registration verification codes using Brevo or Resend HTTP APIs.
 - Dynamic Styling & Dark Mode: A sleek glassmorphic layout supporting a responsive user interface and manual dark mode toggling.
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■■ Tech Stack

- Backend: PHP 8+ (using the Native MongoDB Driver extension)
 - Database: MongoDB Atlas (Cloud NoSQL)
 - CSS Framework: Bootstrap 5 (with Custom Glassmorphism styles)
 - Authentication: Google OAuth 2.0 Client Library & Native Session Handlers
 - Email APIs: Brevo (Sendinblue) API & Resend.com API
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■ Local Setup (using XAMPP)

Prerequisites

1. Install **XAMPP** (with PHP 8+).
2. Install the **MongoDB PHP Extension** (`php_mongodb.dll` for Windows) and add `extension=mongodb` to your `php.ini`.
3. Start Apache in XAMPP.

Installation Steps

1. **Clone the Repository:**

Move the project files to your XAMPP htdocs directory:

```
C:\xampp\htdocs\CampusFind-Pro
```

2. **Configure Environment Variables:**

Create a `.env` file in the root folder (`CampusFind-Pro/.env`) and add your credentials:

```
# MongoDB Atlas Database Credentials
MONGODB_URI="mongodb+srv://<username>:<password>@cluster0.mongodb.net/campusfind_pro"
MONGODB_DB="campusfind_pro"

# Google OAuth Credentials
GOOGLE_CLIENT_ID="your-client-id.apps.googleusercontent.com"
GOOGLE_CLIENT_SECRET="your-client-secret"

# SMTP Backup Mailer
SMTP_HOST="smtp.gmail.com"
SMTP_PORT=465
SMTP_USER="your-email@gmail.com"
SMTP_PASS="your-app-password"
SMTP_SECURE="ssl"

# Brevo.com HTTP API Credentials (Allows sending to any recipient)
BREVO_API_KEY="your-brevo-api-key"
BREVO_SENDER_EMAIL="your-verified-sender@domain.com"
BREVO_SENDER_NAME="CampusFind Pro"

# Resend.com HTTP API Credentials (Fallback)
RESEND_API_KEY="your-resend-api-key"
```

```
RESEND_FROM_EMAIL="onboarding@resend.dev"
```

3. **Seed Database Schema:**

To populate initial default categories (e.g., Electronics, Documents, Keys) and create the administrator user account, open a terminal in the root directory and run the database seeder:

```
php database/seed_mongodb.php
```

- **Default Admin Account:** `admin@campusfindpro.edu`
- **Default Admin Password:** `admin123`

4. **Run the Project:**

Open your browser and navigate to:

```
http://localhost/CampusFind-Pro/
```

■ Cloud Deployment (Render.com)

1. Connect your GitHub repository to a new **Render Web Service**.

2. **Environment Variable Configuration:**

Map the following variables under the **Environment** settings tab on Render:

- `MONGODB_URI`
- `MONGODB_DB`
- `GOOGLE_CLIENT_ID`
- `GOOGLE_CLIENT_SECRET`
- `BREVO_API_KEY`
- `BREVO_SENDER_EMAIL`

3. Click **Save Changes** and deploy the web service. Render will automatically compile and launch your application!

2. Implementation Migration Plan

Implementation Plan - MongoDB Atlas Migration

We will perform a complete database layer migration of CampusFind Pro from MySQL to **MongoDB Atlas (NoSQL)**.

To ensure the project remains portable and runs on XAMPP and Render without requiring Composer, we will utilize PHP's compiled core extension classes: `MongoDB\Driver\Manager`, `MongoDB\Driver\Query`, and `MongoDB\Driver\BulkWrite`.

User Review Required

> **[!IMPORTANT]**

> **MongoDB Driver Activation on XAMPP (Local Machine)**

> To run this locally, you must enable the MongoDB PHP extension in your XAMPP installation:

> 1. Download the DLL file matching your PHP version (PHP 8.2 Thread Safe x64) from PECL: `https://pecl.php.net/package/mongodb`.

> 2. Paste `php_mongodb.dll` into your `C:\xampp\php\ext\` directory.

> 3. Add `extension=php_mongodb` to your `C:\xampp\php\php.ini` file.

> 4. Restart Apache in the XAMPP Control Panel.

>

> *(On Render, this is handled automatically during deployment via our updated Dockerfile).*

> **[!IMPORTANT]**

> MongoDB Atlas Connection URL

> You must create a free cluster on MongoDB Atlas (https://www.mongodb.com/cloud/atlas) and obtain your connection string:

> - Format: mongodb+srv://<username>:<password>@cluster.mongodb.net/?retryWrites=true&w=majority

> - We will declare this string inside config/config.php and support environment variables fallback for Render.

Proposed Changes

1. Dockerfile Update

- [MODIFY] [Dockerfile](file:///C:/Users/Dev/.gemini/antigravity/scratch/CampusFind-Pro/Dockerfile) — Install and enable PECL mongodb driver.

```
RUN pecl install mongodb && docker-php-ext-enable mongodb
```

2. Configuration Setup

- [MODIFY] [config.php](file:///C:/Users/Dev/.gemini/antigravity/scratch/CampusFind-Pro/config/config.php) — Replace MySQL configuration constants with MONGODB_URI and MONGODB_DB.

3. Database Layer Migration

- [MODIFY] [database.php](file:///C:/Users/Dev/.gemini/antigravity/scratch/CampusFind-Pro/config/database.php) — Replace PDO connection with a Singleton wrapper for MongoDB\Driver\Manager. Implement utility helper functions:
 - find(collection, filter, options)
 - findOne(collection, filter)
 - insert(collection, document)
 - update(collection, filter, updateData)
 - delete(collection, filter)

- `count(collection, filter)`

4. Seeding MongoDB database

- [NEW] [seed_mongodb.php](file:///C:/Users/Dev/.gemini/antigravity/scratch/CampusFind-Pro/database/seed_mongodb.php) — A PHP script to seed default settings, lost/found categories, and create the default administrator account in MongoDB Atlas.

5. Controller & Module Rewrites

- [MODIFY] [helpers.php](file:///C:/Users/Dev/.gemini/antigravity/scratch/CampusFind-Pro/includes/helpers.php) — Update `logActivity` and `addNotification` to write directly to MongoDB.
 - [MODIFY] All 15+ backend controllers (login, registration, dashboards, CRUD pages) to use the new MongoDB database helper methods instead of SQL commands.
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Verification Plan

Automated Tests

- Validate PHP syntax compile on updated PHP files:

`php -l filename.php`

Manual Verification

1. Run `database/seed_mongodb.php` via command line to initialize collections on your MongoDB Atlas cluster.
2. Verify collections (users, settings, categories) are created in MongoDB Atlas Collections dashboard.
3. Test signing up and signing in to confirm seamless authentication.

3. Technical Walkthrough & Verification

Walkthrough - CampusFind Pro Lost & Found System

CampusFind Pro is completed. We have established a fully normalized database, OOP core config classes, secure session authentication managers, student dashboards, lost and found report managers, claims verification engines, and an admin analytics panel.

1. Directory Structure Generated

The project is structured under `C:\Users\Dev\gemini\antigravity\scratch\CampusFind-Pro`:

```
CampusFind-Pro/
  ■■■ assets/
  ■ ■■■ css/
  ■ ■ ■■■ style.css # Custom premium Glassmorphism & dark themes stylesheet
  ■ ■■■ js/
  ■ ■■■ app.js # Application core theme toggler, AJAX helpers, Toasts
  ■■■ config/
  ■ ■■■ config.php # Environment globals, directories, DB variables
  ■ ■■■ database.php # Secure PDO database connector (Singleton)
  ■ ■■■ session.php # Hijacking defense, timeouts, role checks, CSRF managers
  ■■■ database/
  ■ ■■■ schema.sql # Normalized schema setup (Seed files included)
  ■■■ includes/
  ■ ■■■ helpers.php # Sanitisation checks, logs/notif creators, flash alerts
  ■ ■■■ navbar.php # Main header template (Dynamic session dropdowns)
  ■ ■■■ footer.php # Bottom links & assets compiler (Chart.js, AOS, QRCode)
  ■■■ auth/
  ■ ■■■ login.php # Authentication page (Checks student status, remembers)
  ■ ■■■ register.php # Signup page (Enforces validations, verification logs)
  ■ ■■■ logout.php # Destroys sessions and cookies safely
  ■ ■■■ verify.php # Simulation verifying email codes
  ■ ■■■ forgot-password.php # Recovery page writing simulated code URL to logs
  ■■■ dashboard/
  ■ ■■■ index.php # Student control center (Pills layout, Chart.js counts)
  ■ ■■■ profile.php # Details adjustments, security inputs, avatar uploads
  ■■■ lost/
  ■ ■■■ report.php # Report lost page (Reward field, image parameters)
  ■ ■■■ edit.php # Modification page (Purges old image file assets)
  ■ ■■■ delete.php # DB deletion & image file cleaning
  ■ ■■■ search.php # Multicriteria search list (With dynamic pagination)
  ■ ■■■ view.php # View registry, client QR Code renderer
  ■■■ found/
  ■ ■■■ report.php # Log found item page
```



```
■ ■■■■ edit.php # Found details modifications
■ ■■■■ delete.php # Found item cleansing
■ ■■■■ search.php # Found item criteria filter
■ ■■■■ view.php # View details, claim submission links, QR Codes
■■■■ claims/
■ ■■■■ submit.php # Submit proof logs (Blocks owner claims, duplicate claims)
■ ■■■■ view.php # Individual claim details & administrative decisions
■ ■■■■ process.php # Transactional approval/rejections (Updates item state)
■■■■ notifications/
■ ■■■■ list.php # Inbox items list
■■■■ uploads/
■ ■■■■ default-avatar.png # Generated default profile avatar illustration
■ ■■■■ default-item.png # Generated default item placeholder illustration
■ ■■■■ mock_emails.log # System output writing mock emails & reset tokens
■■■■ admin/
■■■■ dashboard.php # Moderation dashboard (Resolution rates, month trends)
■■■■ users.php # Students freeze controls & deletes
■■■■ items.php # Registry Union query moderation list
■■■■ claims.php # Pending claims review row overlays
■■■■ reports.php # Export reports manager & log list
■■■■ export.php # Excel rendering table header wrapper & file snapshots
■■■■ settings.php # Timeout & verification checkboxes config saves
■■■■ logs.php # Searchable security audit timeline log table
```

2. Verification Steps Done

- **PHP Compile Check:** Evaluated syntax using XAMPP PHP engine on all 36 files. All files passed with zero compilations warnings or failures.
- **Default Assets Injection:** Generated minimalist vectors `default-avatar.png` and `default-item.png` and saved them in `uploads/` for display when users do not provide their own uploads.

3. How to Initialize and Test Locally

1. **Start Apache & MySQL:** Start Apache and MySQL from XAMPP Control Panel.
2. **Database Initialization:**
 - Open phpMyAdmin (`http://localhost/phpmyadmin/`).
 - Create a database called `campusfind_pro`.
 - Import the database schema from `[schema.sql](file:///C:/Users/Dev/.gemini/antigravity/scratch/CampusFind-Pro/database/schema.sql)`.
3. **Workspace Copy:**

- Copy or symlink the folder `C:\Users\Dev\gemini\antigravity\scratch\CampusFind-Pro` into your local XAMPP `htdocs` directory.

4. **Access Platform:**

- Load the homepage at: `http://localhost/CampusFind-Pro/index.php`.

5. **Pre-seeded Administrative Login:**

- **Email:** `admin@campusfindpro.edu`
- **Password:** `Admin123!`

6. **Testing Verification Logs:**

- Register a student account, then open `[mock_emails.log]`(`file:///C:/Users/Dev/gemini/antigravity/scratch/CampusFind-Pro/uploads/mock_emails.log`) inside the uploads folder to retrieve email confirmation codes or forgot password recovery links.